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MIL-D-45394D
24 September 1991
SUPERSEDING
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28 May 1984

MILITARY SPECIFICATION

DOLLIES, TRAILER CONVERTER: 9 TO 35 TONS PAYLOAD, COMMERCIAL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers trailer converter dollies for use in converting semitrailers to full trailers. Vehicles procured under this specification are commercial items which are warranted by the manufacturer as specified in acquisition documents.

1.2 Classification. The dolly shall be one of the types and sizes shown in table I, as specified (see 6.2).

TABLE I. Dolly classification.

Type	Nomenclature	Size
I	Single axle	9
	Single axle	11
	Single axle	13
II	Tandem axle	20
	Tandem axle	25
	Tandem axle	35

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: AMSTA-URD, Warren, MI 48397-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC-2330

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1.2.1 Size. The size of the dolly shall be determined by its total rated payload capacity as specified in table II.

TABLE II. Size.

Size	Payload capacity (tons)	Payload capacity (pounds)
9	9	18,000
11	11	22,000
13	13	26,000
20	20	40,000
25	25	50,000
35	35	70,000

2. APPLICABLE DOCUMENTS

2.1 Government documents.

* 2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

STANDARDS

FEDERAL

FED-STD-297 - Rustproofing of Commercial (Nontactical) Vehicles.

MILITARY

MIL-STD-209 - Slinging And Tiedown Provisions For Lifting and Tying Down Military Equipment and Related Materials.

MIL-STD-1223 - Non-Tactical Wheeled Vehicles Treatment, Painting, Identification Marking, and Data Plate Standards.

MS 51339 - Coupler, Drawbar, Ring-Offset (Taper Shank).

MS 75020 - Connector, Plug, Electrical-12 Contact, Intervehicular, 28 Volt, Waterproof.

MS 75021 - Connector, Receptacle, Electrical-12 Contact, Intervehicular, 28 Volt, Waterproof.

* (Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, Military Specifications and Standards, Bldg 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

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* 2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

* DEPARTMENT OF DEFENSE (DOD)
Department of Defense Index of Specifications and Standards (DODISS).

(Copies of the DODISS are available on a yearly subscription basis either from the Government Printing Office for hard copy, or microfiche copies are available from the Director, Navy Publication and Printing Service Office, 700 Robbins Avenue, Philadelphia, PA 19111-5093.)

DEPARTMENT OF TRANSPORTATION (DOT)
Federal Motor Carrier Safety Regulations.
Federal Motor Vehicle Safety Standards.

* (Application for copies of DoT publications should reference the Code of Federal Regulations, 49CFR, and the Federal Register, and should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington D.C. 20402.)

* 2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

* AMERICAN WELDING SOCIETY (AWS)
Structural Welding Code, Steel, D1.1

(Applications for copies of AWS publications should be addressed to the American Welding Society, 2501 NW 7th Street, Miami, FL 31025.)

* THE EUROPEAN TYRE AND RIM TECHNICAL ORGANISATION (ETRTO)
Standards Manual

(Application for copies of ETRTO publications should be addressed to the European Tyre and Rim Technical Organisation, 32, Avenue Brugmann, 1060 Brussels, Belgium.)

SAE, INC.
SAE Standards and Recommended Practices
J318 - Air Brake Gladhand Service (Control) and Emergency (Supply) Line Couplers - Trucks, Truck-Tractors, and Trailers (DOD adopted).

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J560	- Seven-Conductor Electrical Connector for Truck-Trailer Jumper Cable.
J697	- Safety Chain of Full Trailers or Converter Dollies.
J700	- Fifth Wheel Kingpin - Commercial Trailers and Semitrailers (DoD adopted).
J849	- Connection and Accessory Locations for Towing Double Trailers and Multi-Axle Trailers.
J1067	- Seven-Conductor Jacketed Cable for Truck and Trailer Connections.
J1292	- Automobile, Truck, Truck-Tractor, Trailer, and Motor Coach Wiring.

(Application for copies should be addressed to SAE, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

THE TIRE AND RIM ASSOCIATION, INC.
Year Book.

* (Application for copies of Tire and Rim Association publications should be addressed to The Tire and Rim Association, Inc., 175 Montrose West Ave., Copley, OH 44321.)

* (Non-Government standards and other publications are normally available from the organizations that prepare or distribute the document. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Standard vehicle and accessories. Except as specified in 3.1.1 through 3.1.1.8, the vehicle, components, assemblies, and accessories to be delivered under the contract shall be standard or optional items, which meet or exceed the requirements of this specification. All vehicle items shall be as represented and rated in the vehicle and equipment manufacturer's technical data. Technical data shall be limited to specifications and technical material, identical to that furnished to the authorized company representatives for selection of vehicle models and components, and shall be available to the engineering offices of the procuring activity, prior to delivery of the items.

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3.1.1 Special requirements. In addition to the standard vehicle and components specified in 3.1, the vehicle shall be furnished with special equipment as specified herein.

3.1.1.1 Treatment and painting. The vehicle shall be treated and painted in accordance with MIL-STD-1223. As specified by the procuring activity for the appropriate military service (see 6.2), the exterior color shall be in accordance with MIL-STD-1223.

3.1.1.2 Markings and data plates. As specified by the procuring activity for the appropriate military service (see 6.2), identification marking and data plates shall be in accordance with MIL-STD-1223. When specified (see 6.2), concealed markings shall be furnished.

* 3.1.1.3 Rustproofing. When specified (see 6.2), the vehicle shall be rustproofed in accordance with FED-STD-297. When specified (see 6.2), tropical rustproofing in accordance with FED-STD-297 shall be furnished.

* 3.1.1.4 Additional data plates. Additional data plates depicting the maximum towing speeds and permissible loads shall be provided. The plates shall indicate the maximum towing speeds with and without payload. Appropriate loading instructions and cautions, as applicable to the manufacturer's design, shall be furnished on the plates. A plate shall be provided on each side of the vehicle in a conspicuous location. The plates shall conform to composition A (class 1 or 2) or composition C of MIL-P-514.

* 3.1.1.5 Lifting and tiedown attachments. When specified (see 6.2), the dolly shall be equipped with lifting and tying down attachments. Lifting and tying down attachments shall conform to type II or type III of MIL-STD-209. In MIL-STD-209G, replace "maximum shipping weight (MSW)" wherever it appears with "curb weight." A transportation plate conforming to composition A (class 1 or 2) or composition C of MIL-P-514 shall be attached to the dolly. The transportation plate shall be inscribed with a diagram showing the lifting attachments and proposed lifting slings, the capacity of each attachment, and the required length and size of each sling cable. A silhouette of the dolly, showing the center of gravity, shall be provided on the transportation plate. Tying down attachments shall be identified by stenciling or other suitable marking. Tiedown markings shall clearly indicate that the attachments are intended for the tiedown of the dolly on the carrier when shipped.

* 3.1.1.6 Lubrication chart. When specified (see 6.2), an illustrated lubrication chart shall be provided. The chart shall direct attention to all lubrication fittings and specify the range and grade of lubricants required for critical temperatures. The chart shall be permanently attached to the dolly in a readily visible location. The chart shall be inscribed on a plate conforming to composition A (class 1 or 2) or composition C of MIL-P-514.

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* 3.1.1.7 Caution plate and instruction plates. Caution and instruction plates shall be furnished and mounted at a conspicuous location on the dolly. The plates shall be of corrosion-resistant material.

3.1.1.8 Lunette. A lunette shall be installed and shall be in accordance with MS 51339.

3.2 General design.

3.2.1 Federal Motor Vehicle Safety Standards. The vehicle shall comply with all Federal Motor Vehicle Safety Standards applicable to the type of dolly furnished and in effect on the date of manufacture.

* 3.2.2 Dissimilar metals. All dissimilar metals used throughout the dolly shall be insulated from one another to prevent galvanic or electrolytic action.

3.2.3 Ratings. Dolly ratings shall be the manufacturer's published ratings. Component and vehicular ratings shall not be raised to meet the requirements of this specification. When published ratings are not available, verification of ratings shall be available to the appropriate engineering office of the procuring activity.

* 3.2.4 Accessibility. The design of the dolly and optional equipment shall permit access for routine servicing and shall permit access for replacement and adjustment of component parts and accessories with minimal disturbance of other components and systems.

* 3.2.5 Prohibited materials. Asbestos materials shall not be used in any form in any part of the vehicle. No item, part or assembly shall contain radioactive materials in which the specific activity is greater than 0.002 microcurie per gram or activity per item equals or exceeds 0.01 microcuries.

3.2.6 Curb weight. The curb weight of the fully equipped dolly shall be the minimum practicable for the service intended.

3.2.7 Payload. The dolly shall be capable of transporting the rated payload specified in table II, with the payload evenly distributed over the area of the fifth wheel. The dolly shall be capable of pulling a gross towing weight equal to twice the payload specified in table II (half the gross towing weight on the dolly, half the gross towing weight on the semitrailer axle(s)).

3.3 Performance. Dolly performance requirements as specified in 3.3.1 through 3.3.4 shall be met by the dolly when fully equipped; completely serviced; coupled to a semitrailer that imposes the rated payload on the dolly fifth wheel and the gross towing weight on the drawbar; and towed by a

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towing vehicle over improved and unimproved hard uneven terrain. The dolly shall show no evidence of failure or malfunction of the dolly, dolly components, attachments, and accessories.

* 3.3.1 Towing speed and trailing ability. When transporting the rated payload (see 1.2.1), while being towed over dry, level, smooth, improved, and prepared hard-surfaced roads at sustained speeds to 55 miles per hour (mph), and over dry, unimproved and gravel roads at speeds up to 20 mph, the semitrailer and dolly shall follow the towing vehicle without exceeding the tracking deviations as specified in Federal Motor Carrier Safety Regulation 393.70(a). When specified (see 6.2), a cambered axle shall be furnished to improve trailing and tracking performance.

3.3.2 Turning ability. The dolly shall be capable of moving through an angle of not less than 55 degrees in relation to the coupled towing vehicle without cramping or damage to the dolly, the towing vehicle, and the coupled semitrailer. The angle shall be measured between the longitudinal axis of the dolly and the longitudinal axis of the towing vehicle at the lunette. The dolly shall be capable of assuming an angle of 90 degrees to the coupled semitrailer without damage to the dolly and semitrailer. Turning ability shall be demonstrated when the dolly is coupled to a 96 inch wide towing vehicle with a pintle 5 inches forward at the rear of the towing vehicle and to a semitrailer having the kingpin located 36 inches from the front of the semitrailer, a width of 96 inches, and square corners. When the semitrailer used in demonstration is furnished without square corners, calculations shall be furnished.

3.3.3 Suspension system. The axle(s), springs, accessories and components of the suspension system shall withstand the imposed loads without evidence of permanent set, part failure or malfunctioning under the operation conditions specified.

3.3.4 Braking ability. The service brakes of the dolly in conjunction with the semitrailer and the towing vehicle shall control, decelerate and stop the vehicle combination in accordance with Federal Motor Carrier Safety Regulation 393.52.

3.4 Dimensions and clearances. The dolly, without payload, uncoupled from the towing vehicle, resting on level ground, shall conform to the dimensions specified in table III.

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TABLE III. Dimensions and clearances.

Dolly size	Dimensions			
	9, 11, 13 and 20	25 and 35	Type I, 9, 11 and 13	Type II, 20, 25 and 35
Overall width over tires	100 inches maximum	100 inches maximum		
Overall length	Minimum practicable	Minimum practicable		
Unladen height of fifth wheel	48 inches ± 1 inch	54 inches ± 1 inch		
Centerline of axle to rear of dolly			30 inches maximum	
Centerline of tandem axle assembly to rear of dolly				60 inches maximum

3.5. Dolly components.

3.5.1 Frame. Main frame members shall be adequately braced with a sufficient number of crossmembers to insure stability. When the dolly is statically loaded with load distributed as specified, the maximum fiber stress in any frame member shall be not greater than 50 percent of the yield strength of the material used. All parts of the frame shall be completely electric arc welded.

* 3.5.2 Suspension. The dolly shall be equipped with a spring suspension system with each component having a rated capacity at least equal to the load imposed, measured at the ground, with the dolly loaded with the payload specified in table II. When specified (see 6.2), a two stage spring suspension system shall be furnished. When specified (see 6.2), a transverse spring suspension system shall be furnished.

3.5.3 Axle. The dolly shall be provided with a single or tandem axle as specified in table I. The axle rating shall be at least equal to the load imposed, measured at the ground, when the dolly is loaded with the payload specified in table II.

3.5.3.1 Oil lubricated axles. The wheel bearings and axle spindles shall be oil lubricated. The oil viscosity shall be in accordance with the manufacturer's recommendations. The hubs shall be fitted with seals. The hub caps shall have a window and shall incorporate provisions for replenishing the oil supply. The hub caps shall incorporate provisions for venting or equivalent method of withstanding pressures without leakage.

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* 3.5.4 Brakes. Full air internal expanding type brakes shall be furnished. The brakes shall conform to Federal Motor Carrier Safety Regulations 393.40, 393.42, 393.43, and 393.45 through 393.52. The dolly brakes shall be capable of applications both with and without a coupled semitrailer. The brake system shall include piping, air reservoirs, automatic slack adjusters, relay emergency valve, four air hose lines permanently attached to the dolly at one end and with SAE J318 gladhands and emergency and service identification tags at their free ends, and all other parts and connections required for a complete brake system. All components of the braking system shall be installed in such a manner as to provide adequate road clearance when traveling over uneven or rough terrain. No part of the brake system shall extend below the bottom of the wheel rims to insure, in case of a flat tire, that the weight of the dolly will be supported by the rim and the flat tire and not be imposed on any component of the braking system. Brake linings shall be of non-asbestos material.

3.5.4.1 Air hoses. All air hoses shall be of the coiled nylon type conforming to Federal Motor Carrier Safety Regulation 393.45(c). Two of the air hose lines shall be of sufficient extended length to connect the dolly to a semitrailer under the turning conditions specified in 3.3.2. Spring supports on the dolly with dummy gladhand connectors to retain the dolly-to-semitrailer hoses when not in use shall be provided. The other two air hose lines shall be of sufficient extended length to connect the dolly to a towing vehicle with the towing vehicle gladhands located at the most extreme location condition allowed by SAE J849. The dolly-to-towing vehicle gladhands shall be provided with spring-loaded cover assemblies.

3.5.5 Lighting. The dolly lights, reflectors and wiring shall conform to Federal Motor Carrier Safety Regulations 393.14(d), 393.23, 393.25 through 393.29 and 393.31 through 393.33. Not less than one stoplight, one taillight and two reflectors shall be provided on the rear. Lights and reflectors shall be mounted in recessed or otherwise guarded locations. All electrical wiring shall conform to SAE J1292. Appropriate wiring, electrical connections and intervehicular electrical cable(s) shall be provided to interconnect the towing vehicle and dolly lighting systems.

3.5.5.1 Twelve volt system. A 12 volt direct current (dc) lighting system conforming to 3.5.5 shall be provided. A trailer electrical receptacle conforming to SAE J560 shall be installed on the forward end of the drawbar. The receptacle shall have a spring loaded cover assembly. Conductors shall be connected and color coded as specified in SAE J560.

3.5.5.1.2 Twelve volt lighting cable. A 7-conductor electrical jumper cable conforming to SAE J1067, with a cable plug at each end conforming to SAE J560, shall be provided. Both plugs shall be equipped with grips for withdrawing from the connector sockets. The cable(s) shall be of sufficient extended length to connect the empty dolly to a towing vehicle with the towing vehicle electrical receptacle located at the most severe condition

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allowed by SAE J849. The length of the cables shall permit turning under the conditions specified in 3.3.2. Stowage provisions having an anti-theft feature and means of hanging the cable, with plugs pointing downward, shall be provided.

3.5.5.2 Independent 24-volt dc system. When specified (see 6.2), a 24-volt dc lighting system, independent of the 12-volt dc lighting system, shall be provided. Separate wiring and bulbs shall be provided for the 24-volt dc system. The system shall conform to 3.5.5, 3.5.5.4, 3.5.5.5 and 3.5.5.6.

3.5.5.3 Interconnected 24-volt dc system. When specified (see 6.2), capability of operating the 12-volt dc system (see 3.5.5) from a towing vehicle equipped with a nominal 24-volt dc electrical system shall be provided by utilizing dropping resistors in the 24 volt to 12 volt lighting circuits. The system shall conform to 3.5.5, 3.5.5.4, 3.5.5.5 and 3.5.5.6.

3.5.5.4 Receptacle for 24-volt dc system. When either an independent or interconnected 24-volt dc system is specified, the front of the dolly shall also be equipped with a 12-contact receptacle and cover conforming to MS 75021, part number MS 75021-1. The 24-volt dc, 12-contact receptacle shall be connected to the lights as follows:

Contact B. Connect to stop lamp (red)
Contact D. Connect to ground (white)
Contact E. Connect to taillamp (brown)

The remaining contacts shall not be connected.

3.5.5.5 Resistors. When an interconnected 24-volt dc system is specified (see 3.5.5.3), the 12-contact receptacle shall be provided with resistance to each circuit to reduce the voltage of the tactical towing vehicle from a nominal 24 volts dc to 12 volts dc. Each circuit resistor shall be selected to reduce the voltage of the tactical truck to within the maximum rated voltage of the dolly electrical components. The resistor assembly shall be located in a protective housing and provided with adequate ventilation or a heat sink to preclude overheating and any damage to resistors, wiring, or adjacent components.

3.5.5.6 Twenty-four volt lighting cable. When a 24-volt system is specified, a 12 conductor cable with connector conforming to MS 75020-1 on one end and to MS 75020-2 on the other end, shall be furnished. The length of the cable shall be as required in 3.5.5.2. Stowage provisions having an anti-theft feature and means for hanging the cable, with plugs pointing downward, shall be provided.

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* 3.5.6 Wheels, rims, tires and tubes. The dolly shall be equipped with dual wheels. Rims and tire ratings shall conform to Tire and Rim Association or European Tyre and Rim Technical Organisation recommendations for the type and size of tires furnished. Except when tube type tires are specified in acquisition documents, multi-piece rims shall not be furnished. Tire and rim sizes shall be the same for all wheels on each dolly. When specified (See 6.2) disc type wheels shall be furnished.

* 3.5.6.1 Tires. Steel belted radial tires shall be furnished. Unless otherwise specified tires shall be tubeless type with highway tread. Unless otherwise specified, low profile or standard profile tires shall be furnished. When specified (see 6.2), low profile tires shall be furnished. The tires shall be of a rated capacity at least equal to the load imposed on each tire, measured at each wheel, at the ground, with the dolly loaded to its specified payload capacity. Tires and tire size designation systems shall conform to the Tire and Rim Association or to the European Tyre and Rim Technical Organisation recommendations.

3.5.6.2 Inner tubes. When tube type tires are specified in acquisition documents, inner tubes shall be of the heavy duty type and shall be proper size for the tire furnished. Tire flaps shall be provided for tube type tires in accordance with Tire and Rim Association recommendations.

* 3.5.7 Drawbar. The drawbar, including its attachments to the dolly and the towing vehicle, shall withstand the drawbar pull specified in table IV without residual deformation. The drawbar shall conform to Federal Motor Carrier Safety Regulation 393.70. When specified (see 6.2), an adjustable length drawbar shall be furnished.

TABLE IV. Drawbar pull.

Size of dolly	Drawbar pull (pounds)
9	45,000
11	50,000
13	55,000
20	90,000
25	115,000
35	160,000

3.5.7.1 Drawbar (hinged-type). Unless otherwise specified (see 3.5.7.2), the drawbar shall be of the hinged-type. (A hinged-type drawbar is available only when dolly is provided with a non-rocking, non-tilting fifth wheel.) The drawbar when in horizontal position shall be 36 inches above ground level. The drawbar shall be capable of being made rigid in the horizontal position and in not less than seven other positions as follows: 8 inches above the horizontal, and six equal positions below the horizontal to within 8 inches above ground level. The dolly shall be capable of being towed with the drawbar in any of the positions specified. During coupling

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and uncoupling operations, and when the dolly is being towed or stowed without payload, the drawbar shall be made rigid. Caution and instruction plates shall be furnished. Means to lubricate the hinged area of the drawbar shall be provided.

* 3.5.7.1.1 Drawbar lock. A drawbar lock shall be provided to maintain the drawbar in each of the eight positions specified. The lock shall be a quick-lock or unlock device. Means shall be provided to preclude the lock from engaging or disengaging during travel operations. A minimum 60 pound breakaway weight at level height is required.

3.5.7.2 Drawbar (rigid type). When specified (see 6.2), the drawbar shall be of the rigid-type. (The rigid-type drawbar is available only when the dolly is provided with a rocking type fifth wheel.) The drawbar, when in the horizontal position, shall be 36 inches above ground level.

* 3.5.7.3 Lunette. A lunette shall be furnished. The lunette shall conform to 3.1.1.8 and shall have a vertical range of adjustment of not less than 4 inches. The lunette shall be induction hardened. When specified (see 6.2), the drawbar eye shall be rubber mounted.

3.5.7.4 Landing gear. Landing gear shall be furnished, located on the drawbar. The landing gear shall be capable of supporting that portion of the imposed load when the dolly is loaded with the specified payload capacity. The landing gear shall have a vertical range of adjustment. The landing gear shall be capable of raising and lowering freely the imposed loads throughout the full range of extension and retraction without evidence of binding or deformation. The landing gear shall be capable of folding into the travel position. Flat landing pads shall be provided. The leg adjusting mechanisms shall be protected from exposure to foreign matter which would impair functional efficiency. Means shall be provided to lubricate the landing gear.

3.5.7.5 Safety chains. Safety chains shall be furnished for the applicable type of drawbar specified herein. Safety chains shall conform to Federal Motor Carrier Safety Regulation 393.70 and SAE J697.

* 3.5.8 Fifth wheel. The fifth wheel shall be 36 inches in diameter, shall be designed for use with an SAE standard fifth wheel kingpin in accordance with SAE J700 and shall conform to Federal Motor Carrier Safety Regulation 393.70. The fifth wheel shall be of the open-throat type coupler plate. The coupler shall be attached in such a manner as to permit removal and replacement, and shall be mounted directly over the centerline of the axle on single axle dollies. The fifth wheel shall be capable of being uncoupled by the operator while standing on the street side of the vehicle. Uncoupling action shall be protected by a secondary lock, preventing movement of the uncoupling lever until the secondary lock is manually released.

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3.5.8.1 Rigid-type. A rigid-type fifth wheel shall be furnished when the dolly is equipped with a hinged-type drawbar as specified in 3.5.7.1.

3.5.8.2 Rocking-type. A rocking type fifth wheel shall be furnished when the dolly is equipped with a rigid-type drawbar as specified in 3.5.7.2).

3.6 Lubrication. All moving parts normally requiring lubrication shall be provided with suitable means for such lubrication.

3.6.1 Lubrication fittings. Lubrication fittings shall be standard commercial type with a surface-check ball.

3.7 Stowage compartment. A stowage compartment shall be installed in a readily accessible location on the dolly. The compartment shall have sufficient capacity to stow the lighting cable(s) (see 3.5.5.1.2 and 3.5.5.6). The compartment shall be equipped with a flanged, hinged lid, a handle for opening and closing and a spring-loaded latch and hasp for locking with a 1 3/4-inch padlock. Drain holes shall be provided in the bottom of the compartment, designed to preclude road splash from entering the compartment.

3.8 Servicing and adjusting. Prior to acceptance by the Government inspector, the contractor shall service and adjust each dolly for operational use including at least the following: adjustment of electrical and brake systems, inflation of all tires, and complete lubrication with grades of lubricants recommended for the ambient air temperature at the delivery point.

* 3.9 Workmanship. Defective components or parts and assemblies which have been repaired or modified to overcome deficiencies shall not be furnished. Welded, bolted, and riveted construction utilized shall be in accordance with the highest standards of the industry. Welding shall conform to the American Welding Society's Structural Welding Code, Steel D1.1 and, when specified (see 6.2), to the requirements of MIL-STD-1261 Class 2.

* 3.9.1 Welding. The surface of parts to be welded shall be free of rust, scale, paint, grease and other foreign matter. Welding rods shall be compatible to the metal being used. Welding shall not be employed as a repair measure for defective parts. Weld design and application shall be such as to provide adequate transference of maximum design stress through the base metal juncture. Fillet welds shall be provided when necessary to reduce stress concentration.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements (examinations, and tests) as specified herein. Except as

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otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility for ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Government verification. Quality assurance operations performed by the contractor will be subject to Government verification at unscheduled intervals. Verification will consist of observation of the operations to determine that practices, methods, and procedures of the contractor's inspection are being properly applied. Failure of the contractor to promptly correct product deficiencies discovered shall be cause for suspension of acceptance until correction has been made or until conformance of product to specification criteria has been demonstrated.

4.3 First production vehicle inspection. The first production vehicle of each type (see 1.2) produced under the contract shall be inspected by the contractor at his plant under the direction and in the presence of Government representatives. The purpose of the inspection shall be to determine vehicle conformance to the contract. Acceptance of the first production vehicle shall not constitute a waiver by the Government of its rights under the provisions of the contract.

4.3.1 Weight and payload capacity. The first production vehicle of each type shall be weighed to determine curb weight. The contractor shall furnish certification that the dolly convertor will accept the imposed loads on the kingpin, dolly frame, suspension, axle(s) and tires and on the drawbar in accordance with contract requirements.

* 4.3.2 Road test. To determine conformance to 3.3 through 3.4, the first production dolly of each type, with rated payload, shall be road tested. The dolly coupled to a semitrailer and with the drawbar attached to a towing vehicle shall be driven not less than 100 miles under the conditions specified in 3.3. At least 20 percent of the distance shall be over dry unimproved and gravel roads at speeds up to 20 mph. The dolly in conjunction

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with the semitrailer and the towing vehicle shall be subjected to not less than three sudden stops at a speed not less than 20 mph on a substantially level, dry, smooth surface, free of loose material. The vehicle combination shall be observed for the specified stopping distance. During the turning ability operation, the dolly shall be observed for moving through an angle of not less than 55 degrees in relation to the coupled towing vehicle without cramping or damage to the dolly, towing vehicle and coupled semitrailer. The vehicle combination shall be observed for damage to the dolly and semitrailer when assuming an angle of 90 degrees to the coupled semitrailer. During the towing speed and trailing ability operation, the semitrailer shall be observed for conformance to the specified tracking deviations. Upon completion of the road test, all equipment, components and accessories shall be examined for damage and the dolly shall be examined for leakage (air and lubricant).

4.3.3 Electrical test. An electrical test shall be conducted on the first production dolly to determine that all lights are functioning. In addition, when an interconnected 24-volt system is furnished, the circuits of the 12-contact receptacle shall be energized with not less than 24 volts dc regulated voltage for a time sufficient to stabilize the temperature of the resistor assembly for not less than 30 minutes in an ambient temperature of not less than 77°F. After the test, the resistor assembly, including resistors, wiring and adjacent components shall be examined for evidence of overheating, deterioration or damage.

4.3.4 Production sample. Upon acceptance of the first production vehicle of each type, it shall remain at the manufacturing facility as a production sample, and shall be the last vehicle(s) shipped on the contract. The contractor shall maintain the vehicle(s) in a serviceable condition for the duration of the contract.

* 4.3.5 Failure. Failure of the first production vehicle(s) to meet requirements of the contract shall be cause for the Government to refuse acceptance of all vehicles under contract until the corrective action has been taken.

4.4 Inspection of production vehicle. Each production dolly shall be subjected to dimensional inspection specified in section 3. The contractor's inspection system, as a minimum, shall assure that the vehicle conforms to the dimensional and performance requirements and the vehicle is capable of meeting the requirements specified herein. For each vehicle under contract, the contractor shall make available to the Government, at the point of final acceptance, records acceptable to the Government indicating that the servicing and adjusting required by 3.8 have been accomplished.

5. PACKAGING

5.1 Vehicle processing. The vehicle shall be processed for shipment, from manufacturer's plant to initial receiving activity, in accordance with the manufacturer's standard commercial practice.

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6. NOTES

6.1 Intended use. The vehicles covered by this specification are intended for general non-tactical use by the Government, for use by the U.S. Armed Forces, for coupling of semitrailers and converting semitrailers to full trailers.

* 6.2 Ordering data. Acquisition documents should specify the following:
(a) Title, number, and date of this specification.
(b) Type and size of dolly required (see 1.2).
* (c) Issue of DODISS to be cited in the solicitation and if required, the specific issue of individual documents referenced (See 2.1.1 and 2.2).
(d) Identification of appropriate service for painting and marking (see 3.1.1.1).
(e) Marking and data plates for appropriate service (see 3.1.1.2).
(f) Rustproofing, if required (see 3.1.1.3).
(g) Lifting and tiedown provisions, if required (see 3.1.1.5).
(h) Lubrication chart, if required (see 3.1.1.6).
* (i) A cambered axle, if required (see 3.3.1).
* (j) A two stage spring assembly, if required (see 3.5.2).
* (k) A transverse spring suspension, if required (see 3.5.2).
(l) Independent 24-volt dc system, if required (see 3.5.5.2).
(m) Interconnected 24-volt dc system, if required (see 3.5.5.3).
(n) Disc type wheels, if required (see 3.5.6).
(o) Low profile tires, if required (see 3.5.6.1).
* (p) An adjustable length drawbar, if required (see 3.5.7).
(q) Rigid type drawbar, if required (see 3.5.7.2).
* (r) Rubber mounted drawbar, if required (see 3.5.7.3).

* 6.3 Subject term (key word) listing.
Dolly components
Drawbar
Fifth wheel
Landing gear
Lunette

6.4 Identification of changes. The margins of this specification are marked with asterisks (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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